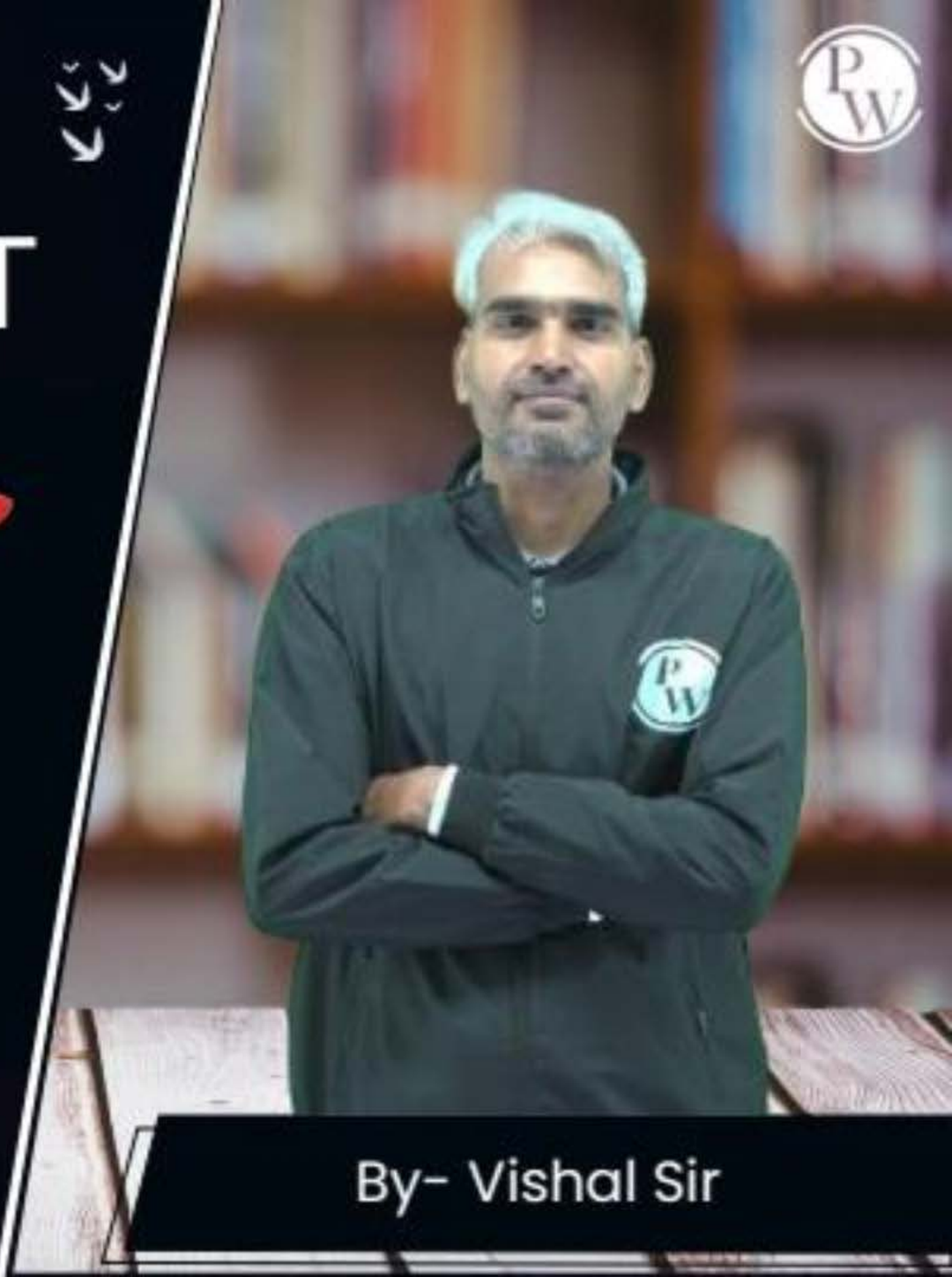
Computer Science & IT



Database Management System

Query Languages

Lecture No. 04



By- Vishal Sir

Recap of Previous Lecture



Topic Practice questions of Relational Algebra



Slide

Topics to be Covered

Topic	Practice questions of Relational Algebra



Slide

Consider the following relational tables:

- Supplier (Sid, Sname, Rating)
- Parts (Pid, Pname, Color)
- Catalog (Sid, Pid, Cost)

Suppliers

Sid	Sname	Rating
S_1	A	3
S_2	A	5
S_3	B	7
S_4	C	0

Parts

Pid	Pname	Color
P_1	XYZ	Red
P_2	PQR	Green
P_3	XYZ	Red

Catalog

Sid	Pid	Cost
S_1	P_1	10
S_2	P_2	20
S_3	P_3	30

* Retrieve Sids of suppliers who supplied at least one part.

π_{sid} (Catalog)

$\Rightarrow$ o/p

Sid
s_1
s_2
s_3

* Retrieve Sids of all the suppliers

π_{sid} (Supplier)

$\Rightarrow$ o/p

Sid
s_1
s_2
s_3
s_4

Q:- Retriere Sids of the suppliers whose rating is more than 5

$$\pi_{Sid}\left(\sigma_{Rating>5}(Supplier)\right)$$

o/p =

Sid
S_3

o/p =

Sid	Sname	Rating
S_3	B	7

Q. Retrieve Sids of Suppliers who supplied some red color parts.

Catalog

Sid	Pid	Cost
S_1	P_1	10
S_2	P_2	20
S_3	P_3	30

Parts

Pid	Pname	Color
P_1	XYZ	Red
P_2	PQR	Green
P_3	XYZ	Red

① $\pi_{C.Sid}\left(\sigma_{(C.Pid = P.Pid \wedge P.Color = 'Red')}\left(\rho_C(Catalog) \times \rho_P(Parts)\right)\right)$

⇓

o/p =

Sid
S_1
S_3

C.Sid	C.Pid	C.Cost	P.Pid	P.name	P.Color
S_1	P_1	10	P_1	XYZ	Red
S_1	P_1	10	P_2	PQR	Green
S_1	P_1	10	P_3	XYZ	Red
S_2	P_2	20	P_1	XYZ	Red
S_2	P_2	20	P_2	PQR	Green
S_2	P_2	20	P_3	XYZ	Red
S_3	P_3	30	P_1	XYZ	Red
S_3	P_3	30	P_2	PQR	Green
S_3	P_3	30	P_3	XYZ	Red

Catalog

Sid	Pid	Cost
S_1	P_1	10
S_2	P_2	20
S_3	P_3	30

Parts

Pid	Pname	Color
P_1	xyz	Red
P_2	PQR	Green
P_3	xyz	Red

(2) $\pi_{Sid}\left(\sigma_{Color='Red'}(C \bowtie P)\right)$

C.Sid	C.Pid	C.Cost	P.Pid	P.name	P.Color
S_1	P_1	10	P_1	xyz	Red
S_1	P_1	10	P_2	PQR	Green
S_1	P_1	10	P_3	xyz	Red
S_2	P_2	20	P_1	xyz	Red
S_2	P_2	20	P_2	PQR	Green
S_2	P_2	20	P_3	xyz	Red
S_3	P_3	30	P_1	xyz	Red
S_3	P_3	30	P_2	PQR	Green
S_3	P_3	30	P_3	xyz	Red

Catalog

Sid	Pid	Cost
S_1	P_1	10
S_2	P_2	20
S_3	P_3	30

Parts

Pid	Pname	Color
P_1	XYZ	Red
P_2	PQR	Green
P_3	XYZ	Red

(3) $\pi_{Catalog.Sid}\left(\sigma_{Catalog.Pid = Parts.Pid}\left(Catalog \times \left(\sigma_{Color='Red'}(Parts)\right)\right)\right)$

(4) $\pi_{Sid}\left(Catalog \bowtie \left(\sigma_{Color='Red'}(Parts)\right)\right)$

Q. Retrieve Sids of Suppliers who supplied some red color parts.

1000 1000

① $\pi_{C.sid}\left(\sigma_{C.Pid=P.Pid \wedge P.Color='Red'}(C \times P)\right)$ — $C \times P$: 1000000

② $\pi_{sid}\left(\sigma_{Color='Red'}(C \bowtie P)\right)$

} Less efficient

50,000

③ $\pi_{C.sid}\left\{\sigma_{C.Pid=P.Pid}\left(C \times \left(\sigma_{Color='Red'}(P)\right)\right)\right\}$ — C: 1000; $\sigma_{Color='Red'}(P)$: 50

④ $\pi_{sid}\left\{C \bowtie \left(\sigma_{Color='Red'}(P)\right)\right\}$

} More efficient

#Q. Retrieve Sid of the suppliers who have supplied some Red or some Green color parts.

Catalog

Sid	Pid	Cost
S_1	P_1	10
S_2	P_2	20
S_3	P_3	30

Parts

Pid	Pname	Color
P_1	XYZ	Red
P_2	PQR	Green
P_3	XYZ	Red

C.Sid	C.Pid	C.Cost	P.Pid	P.name	P.Color
S_1	P_1	10	P_1	XYZ	Red
S_1	P_1	10	P_2	PQR	Green
S_1	P_1	10	P_3	XYZ	Red
S_2	P_2	20	P_1	XYZ	Red
S_2	P_2	20	P_2	PQR	Green
S_2	P_2	20	P_3	XYZ	Red
S_3	P_3	30	P_1	XYZ	Red
S_3	P_3	30	P_2	PQR	Green
S_3	P_3	30	P_3	XYZ	Red

Q. Retrieve Sids of Supplies who supplied some red or some green Color Parts.

① $\pi_{C.sid}\left(\sigma_{(C.Pid=P.Pid \wedge (P.Color='Red' \vee P.Color='Green'))}(C \times P)\right)$

② $\pi_{sid}\left(\sigma_{(Color='Red' \vee Color='Green')}(C \bowtie P)\right)$

③ $\pi_{C.sid}\left\{\sigma_{C.Pid=P.Pid}\left(C \times \left(\sigma_{(Color='Red' \vee Color='Green')}(P)\right)\right)\right\}$

④ $\pi_{sid}\left\{C \bowtie \left(\sigma_{(Color='Red' \vee Color='Green')}(P)\right)\right\}$

#Q. Retrieve Sid of the suppliers who have supplied some Red and some Green color parts.

Q. Retrieve Sids of Suppliers who supplied some red and some green Color Parts.

① $\pi_{C.sid}\left(\sigma_{(C.Pid = P.Pid \wedge (P.Color = 'Red' \wedge P.Color = 'Green'))}(C \times P)\right)$

② $\pi_{sid}\left(\sigma_{(Color = 'Red' \wedge Color = 'Green')}(C \bowtie P)\right)$

③ $\pi_{C.sid}\left\{\sigma_{C.Pid = P.Pid}\left(C \times \left(\sigma_{(Color = 'Red' \wedge Color = 'Green')}(P)\right)\right)\right\}$

④ $\pi_{sid}\left\{C \bowtie \left(\sigma_{(Color = 'Red' \wedge Color = 'Green')}(P)\right)\right\}$

All are wrong.

⇓

All four queries will always produce Empty relation

⇓

In a single tuple, Color will either be 'Red' or 'Green' but Can never be both

Q. Retrieve Sids of Suppliers who supplied some red and some green color parts

① $\pi_{C.sid}\left(\sigma_{C.Pid=P.Pid \wedge P.Color='Red'}(C \times P)\right) \cap \pi_{C.sid}\left(\sigma_{C.Pid=P.Pid \wedge P.Color='Green'}(C \times P)\right)$

② $\pi_{sid}\left(\sigma_{Color='Red'}(C \bowtie P)\right) \cap \pi_{sid}\left(\sigma_{Color='Green'}(C \bowtie P)\right)$

Q. Retrieve Sids of Suppliers who supplied some red and some green color parts

③ $\pi_{C.Sid}\left\{\sigma_{C.Pid = P.Pid}\left(C \times \left(\sigma_{Color='Red'}(P)\right)\right)\right\} \cap \pi_{C.Sid}\left\{\sigma_{C.Pid = P.Pid}\left(C \times \left(\sigma_{Color='Green'}(P)\right)\right)\right\}$

④ $\pi_{Sid}\left\{C \bowtie \left(\sigma_{Color='Red'}(P)\right)\right\} \cap \pi_{Sid}\left\{C \bowtie \left(\sigma_{Color='Green'}(P)\right)\right\}$

Q. Retrieve Sids of Supplies who supplied some red or some green Color parts

① $\pi_{C.sid}\left(\sigma_{C.Pid=P.Pid \wedge P.Color='Red'}(C \times P)\right) \cup \pi_{C.sid}\left(\sigma_{C.Pid=P.Pid \wedge P.Color='Green'}(C \times P)\right)$

② $\pi_{sid}\left(\sigma_{Color='Red'}(C \bowtie P)\right) \cup \pi_{sid}\left(\sigma_{Color='Green'}(C \bowtie P)\right)$

Q. Retrieve Sids of Suppliers who supplied some red or some green Color Parts

③ $\pi_{C.Sid}\left\{\sigma_{C.Pid = P.Pid}\left(C \times \left(\sigma_{Color='Red'}(P)\right)\right)\right\} \cup \pi_{C.Sid}\left\{\sigma_{C.Pid = P.Pid}\left(C \times \left(\sigma_{Color='Green'}(P)\right)\right)\right\}$

④ $\pi_{Sid}\left\{C \bowtie \left(\sigma_{Color='Red'}(P)\right)\right\} \cup \pi_{Sid}\left\{C \bowtie \left(\sigma_{Color='Green'}(P)\right)\right\}$

- Another approach to retrieve Sids of Suppliers who supplied some red and some green Color parts.

Catalog

Sid	Pid	Cost
S_1	P_1	10
S_2	P_2	20
S_3	P_3	30
S_1	P_2	30

Parts

Pid	Pname	Color
P_1	xyz	Red
P_2	PQR	Green
P_3	xyz	Red

$(C_1 \times P_1) \times (C_2 \times P_2)$

C_1.Sid	C_1.Pid	C_1.Cost	P_1.Pid	P_1.Pname	P_1.Color
S_1	P_1	10	P_1	xyz	Red
S_1	P_1	10	P_2	PQR	Green
S_1	P_1	10	P_3	xyz	Red
S_2	P_2	20	P_1	xyz	Red
S_2	P_2	20	P_2	PQR	Green
S_2	P_2	20	P_3	xyz	Red
S_3	P_3	30	P_1	xyz	Red
S_3	P_3	30	P_2	PQR	Green
S_3	P_3	30	P_3	xyz	Red

C_2.Sid	C_2.Pid	C_2.Cost	P_2.Pid	P_2.Pname	P_2.Color
S_1	P_1	10	P_1	xyz	Red
S_1	P_2	30	P_2	PQR	Green
S_1	P_1	10	P_3	xyz	Red
S_2	P_2	20	P_1	xyz	Red
S_2	P_2	20	P_2	PQR	Green
S_2	P_2	20	P_3	xyz	Red
S_3	P_3	30	P_1	xyz	Red
S_3	P_3	30	P_2	PQR	Green
S_3	P_3	30	P_3	xyz	Red

- Another approach to retrieve Sids of Suppliers who supplied some red and some green Color parts.

$$\pi_{C_1.Sid}\Big\{\sigma\big(C_1.Sid = C_2.Sid \wedge (C_1.Pid = P_1.Pid \wedge P_1.Color = 'Red') \wedge (C_2.Pid = P_2.Pid \wedge P_2.Color = 'Green')\big)\{(C_1 \times P_1) \times (C_2 \times P_2)\}\Big\}$$

We are using $C_1 \times P_1$ w.r.t 'Red' Color parts

We are using $C_2 \times P_2$ w.r.t. Green Color parts.

#Q. Retrieve Sid of the suppliers who have supplied all parts.

Catalog (Sid, Pid, Cost)

Parts (Pid, Pname, Color)

We are looking for Sids in Catalog table that are associated with all Pids of Parts table

$\therefore \quad \pi_{Sid,Pid}(C) \div \pi_{Pid}(P)$

$\equiv \quad \left[\pi_{Sid}(C) - \pi_{Sid}\left[(\pi_{Sid}(C) \times \pi_{Pid}(P)) - \pi_{Sid,Pid}(C) \right] \right]$

#Q. Retrieve Sid of the suppliers who have supplied all Red color parts.

* We are looking for the Sids of the supplies that are associated with all Pids of red Color Parts

$$\pi_{Sid, Pid}(C) \div \pi_{Pid}(\sigma_{Color='Red'}(P))$$

$$\equiv \left[\pi_{Sid}(C) - \pi_{Sid}\left[\left(\pi_{Sid}(C) \times \pi_{Pid}(\sigma_{Color='Red}(P))\right) - \pi_{Sid,Pid}(C)\right]\right]$$

For the remaining queries Conside the Pollowing Catalog relation

Catalog

Sid	Pid	Cost
S_1	P_1	10
S_1	P_2	30
S_2	P_2	30
S_3	P_2	20
S_3	P_3	30

#Q. Retrieve Sid of the suppliers who have supplied at least two parts.

Query:- Retrieve Sids of the supplies who have supplied at least two parts

Catalog (C_1)

Sid	Pid	Cost
S_1	P_1	10
S_1	P_2	30
S_2	P_2	30
S_3	P_2	20
S_3	P_3	30

Catalog (C_2)

Sid	Pid	Cost
S_1	P_1	10
S_1	P_2	30
S_2	P_2	30
S_3	P_2	20
S_3	P_3	30

we need to compare tuples of Catalog table with all other tuples of Catalog table

$$\pi_{C_1.Sid}\left(\sigma_{(C_1.Sid = C_2.Sid \wedge C_1.Pid != C_2.Pid)}\left(C_1 \times C_2\right)\right)$$

#Q. Retrieve Sid of the suppliers who have supplied exactly one part.

Suppliers who supplied exactly one Part = (Suppliers who have supplied at least one part) − (Suppliers who have supplied at least two parts)

1, 2, 3, 4, ... — 2, 3, 4, 5 ...

$$= \pi_{Sid}(C) - \pi_{Sid}\left(\sigma_{(C_1.Sid = C_2.Sid \wedge C_1.Pid <> C_2.Pid)}(C_1 \times C_2)\right)$$

#Q. Retrieve Sid of the suppliers who have supplied at most one part.

(0 or 1)

At most one = Any no. of Parts − At least two Parts

{0, 1, 2, 3, ...} − {2, 3, 4, ...}

we also need the Sids of Suppliers who supplied '0' parts

↓

Info about such suppliers is present only in Supplier table.

$$= \left[\pi_{Sid}(Supplier) - \pi_{Sid}\left(\sigma_{(C_1.sid = C_2.Sid} (C_1 \times C_2) \right) \right]$$

$C_1.Pid != C_2.Pid$

#Q. Retrieve Sid of the suppliers who have supplied at least three parts.

$$\Pi_{C_1.Sid}\Big(\sigma_{(C_1.Sid = C_2.Sid \wedge C_2.Sid = C_3.Sid \wedge C_1.Sid = C_3.Sid)}(C_1 \times C_2 \times C_3)\Big)$$

$$(C_1.Pid \neq C_2.Pid \wedge C_2.Pid \neq C_3.Pid \wedge C_1.Pid \neq C_3.Pid)$$

This Condn is optional

w.r.t. Not equal this Condn is Mandatory

we want to Compare each tuple of the Catalog with two more tuples at a time.

We need to perform $C_1 \times C_2 \times C_3$

#Q. Retrieve Sid of the suppliers who have supplied most expensive parts.

i.e, Suppliers who have supplied some Parts at maximum Cost

Tuples of the Catalog table with maximum Cost = All tuples from Catalog table — tuples from Catalog table in which Cost is not maximum

* Retrieve Sids of the suppliers who supplied some parts at maximum cost

Catalog

Sid	Pid	Cost
S_1	P_1	10
S_1	P_2	30
S_2	P_2	30
S_3	P_2	20
S_3	P_3	30

$$\pi_{C_1.Sid,\ C_1.Pid,\ C_1.Cost}\left(\sigma_{C_1.Cost < C_2.Cost}\left(C_1 \times C_2\right)\right)$$

O/P =

C_1.Sid	C_1.Pid	C_1.Cost
S_1	P_1	10
S_3	P_2	20

O/P is the tuples from the Catalog table in which cost is not maximum.

$C_1 \times C_2$



Query: Retrieve Sids of the suppliers with maximum cost

$$\equiv \pi_{Sid}\left[(Catalog) - \pi_{C_1.Sid, C_1.Pid, C_1.Cost}\left(\sigma_{C_1.Cost < C_2.Cost}(C_1 \times C_2)\right)\right]$$

all tuples of Catalog

tuples from Catalog in which Cost is not maximum

Sids of the Suppliers, who have supplied some Parts at Maximum Cost

it results the tuples from the catalog in which cost is maximum

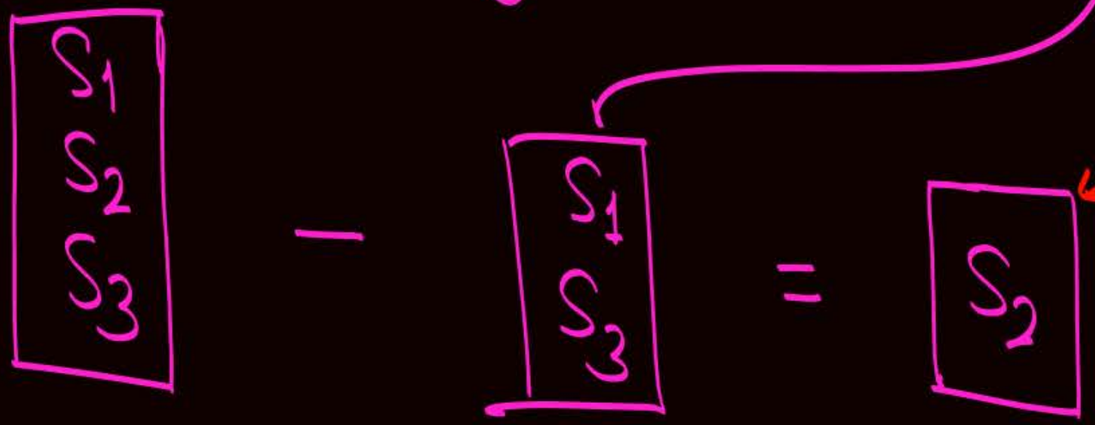
O/P:-

Sid
S1
S2
S3

Query: Retrieve Sids of the suppliers with maximum cost

$$\pi_{Sid}(Catalog) - \pi_{C_1.Sid}\left(\sigma_{C_1.Cost < C_2.Cost}(C_1 \times C_2)\right)$$

o/p w.r.t. given catalog table will be



This query need not produce the correct result.

Q:- Retrieve Sids of suppliers who supplied some red color parts

$$\pi_{C.Sid}\left(\sigma_{C.Pid = P.Pid \,\wedge\, P.Color = 'Red'}\left(C \times P\right)\right)$$

Q:- Retrieve Names of suppliers who supplied some red color parts

$$\pi_{S.Sname}\left(\sigma_{(C.Pid = P.Pid \,\wedge\, P.Color = 'Red' \,\wedge\, C.Sid = S.Sid)}\left(C \times P \times S\right)\right)$$



2 mins Summary

Topic Practice questions of Relational Algebra

Slide

THANK - YOU